

SAT 10

Math Module 3

Question # ID

SAT10 M 3.1 43744269

An airplane descends from an altitude of 9,500 feet to 5,000 feet at a constant rate of 400 feet per minute. What type of function best models the relationship between the descending airplane's altitude and time?

- A. Decreasing exponential
- B. Decreasing linear
- C. Increasing exponential
- D. Increasing linear

SAT10 M 3.2 a10ma302

Line k is defined by $y = \frac{17}{7}x + 4$. Line j is parallel to line k in the xy -plane. What is the slope of line j ?

(A) $\frac{7}{17}$

(B) $\frac{17}{7}$

(C) 4

(D) 17

SAT10 M 3.3 3e9eaffc

Caleb used juice to make popsicles. The function $f(x) = -5x + 30$ approximates the volume, in fluid ounces, of juice Caleb had remaining after making x popsicles. Which statement is the best interpretation of the y -intercept of the graph of $y = f(x)$ in the xy -plane in this context?

- A. Caleb used approximately 5 fluid ounces of juice for each popsicle.
- B. Caleb had approximately 5 fluid ounces of juice when he began to make the popsicles.
- C. Caleb had approximately 30 fluid ounces of juice when he began to make the popsicles.
- D. Caleb used approximately 30 fluid ounces of juice for each popsicle.

SAT 10

Math Module 3

Question # ID
SAT10 M 3.4 a10ma304

A physics class is planning an experiment about a toy rocket. The equation $y = -16(x - 5.6)^2 + 502$ gives the estimated height y , in feet, of the toy rocket x seconds after it is launched into the air. Which of the following is the best interpretation of the vertex of the graph of the equation in the xy -plane?

- (A) This toy rocket reaches an estimated maximum height of 502 feet 16 seconds after it is launched into the air.
- (B) This toy rocket reaches an estimated maximum height of 502 feet 5.6 seconds after it is launched into the air.
- (C) This toy rocket reaches an estimated maximum height of 16 feet 502 seconds after it is launched into the air.
- (D) This toy rocket reaches an estimated maximum height of 5.6 feet 502 seconds after it is launched into the air.

SAT10 M 3.5 a10ma305

The function f is defined by $f(x) = 4x + k(x - 1)$, where k is a constant, and $f(5) = 32$. What is the value of $f(10)$?

SAT10 M 3.6 0bb39de4

Triangles ABC and DEF are congruent, where A corresponds to D , and B and E are right angles. The measure of angle A is 18° . What is the measure of angle F ?

- A. 18°
- B. 72°
- C. 90°
- D. 162°

SAT 10

Math Module 3

Question # **ID**
SAT10 M 3.7 541bef2f

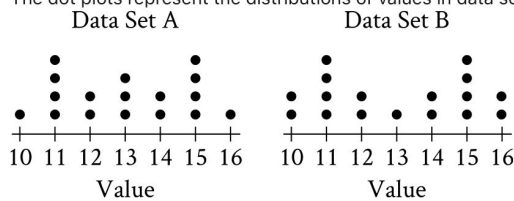
$$\begin{aligned} y &\leq x + 7 \\ y &\geq -2x - 1 \end{aligned}$$

Which point (x, y) is a solution to the given system of inequalities in the xy -plane?

- A. $(-14, 0)$
- B. $(0, -14)$
- C. $(0, 14)$
- D. $(14, 0)$

SAT10 M 3.8 d65b9a87

The dot plots represent the distributions of values in data sets A and B.



Which of the following statements must be true?

- I. The median of data set A is equal to the median of data set B.
 - II. The standard deviation of data set A is equal to the standard deviation of data set B.
- A. I only
 - B. II only
 - C. I and II
 - D. Neither I nor II

SAT10 M 3.9 752055d1

A scientist initially measures 12,000 bacteria in a growth medium. 4 hours later, the scientist measures 24,000 bacteria. Assuming exponential growth, the formula $P = C(2)^{rt}$ gives the number of bacteria in the growth medium, where r and C are constants and P is the number of bacteria t hours after the initial measurement. What is the value of r ?

- A. $\frac{1}{12,000}$
- B. $\frac{1}{4}$
- C. 4
- D. 12,000

SAT10 M 3.10 ba8ca563

A cube has a volume of 474,552 cubic units. What is the surface area, in square units, of the cube?

SAT 10

Math Module 3

Question # ID
SAT10 M 3.11 a10ma311

$$\begin{aligned}2(8x) + 4(7y) &= 12 \\ -2(8x) + 4(7y) &= 12\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of $8x + 7y$?

SAT10 M 3.12 c7c6445f

A certain town has an area of 4.36 square miles. What is the area, in square yards, of this town?
(1 mile = 1,760 yards)

- A. 404
- B. 7,674
- C. 710,459
- D. 13,505,536

SAT10 M 3.13 a10ma313

Triangles PQR and LMN are graphed in the xy -plane. Triangle PQR has vertices P , Q , and R at $(4, 5)$, $(4, 7)$, and $(6, 5)$, respectively. Triangle LMN has vertices L , M , and N at $(4, 5)$, $(4, 7 + k)$, and $(6 + k, 5)$, respectively, where k is a positive constant. If the measure of $\angle Q$ is t° , what is the measure of $\angle N$?

(A) $(90 - (t - k))^\circ$

(B) $(90 - (t + k))^\circ$

(C) $(90 - t)^\circ$

(D) $(90 + k)^\circ$

SAT10 M 3.14 ee7b1de1

A small business owner budgets \$2,200 to purchase candles. The owner must purchase a minimum of 200 candles to maintain the discounted pricing. If the owner pays \$4.90 per candle to purchase small candles and \$11.60 per candle to purchase large candles, what is the maximum number of large candles the owner can purchase to stay within the budget and maintain the discounted pricing?

SAT 10
Math Module 3

Question # **ID**
SAT10 M 3.15 ae041e52

A square is inscribed in a circle. The radius of the circle is $\frac{20\sqrt{2}}{2}$ inches. What is the side length, in inches, of the square?

- A. 20
- B. $\frac{20\sqrt{2}}{2}$
- C. $20\sqrt{2}$
- D. 40

SAT10 M 3.16 9d159400

Which of the following equations represents a circle in the xy -plane that intersects the y -axis at exactly one point?

- A. $(x - 8)^2 + (y - 8)^2 = 16$
- B. $(x - 8)^2 + (y - 4)^2 = 16$
- C. $(x - 4)^2 + (y - 9)^2 = 16$
- D. $x^2 + (y - 9)^2 = 16$

SAT10 M 3.17 5e08a055

$$y = 6x + 18$$

One of the equations in a system of two linear equations is given. The system has no solution. Which equation could be the second equation in the system?

- A. $-6x + y = 18$
- B. $-6x + y = 22$
- C. $-12x + y = 36$
- D. $-12x + y = 18$

SAT10 M 3.18 20291f47

Which expression is equivalent to $\frac{y+12}{x-8} + \frac{y(x-8)}{x^2y-8xy}$?

- A. $\frac{xy+y+4}{x^3y-16x^2y+64xy}$
- B. $\frac{xy+9y+12}{x^2y-8xy+x-8}$
- C. $\frac{xy^2+13xy-8y}{x^2y-8xy}$
- D. $\frac{xy^2+13xy-8y}{x^3y-16x^2y+64xy}$

SAT 10

Math Module 3

Question # ID
SAT10 M 3.19 a10ma319

$$\frac{20}{p} = \frac{20}{q} - \frac{20}{r} - \frac{20}{s}$$

The given equation relates the positive variables p , q , r , and s . Which of the following is equivalent to q ?

(A) $p + r + s$

(B) $20(p + r + s)$

(C) $\frac{prs}{pr+ps+rs}$

(D) $\frac{prs}{20p+20r+20s}$

SAT10 M 3.20 03ff48d2

$$x(kx - 56) = -16$$

In the given equation, k is an integer constant. If the equation has no real solution, what is the least possible value of k ?

SAT10 M 3.21 75012ee7

$$\begin{aligned} 2x + 3y &= 7 \\ 10x + 15y &= 35 \end{aligned}$$

For each real number r , which of the following points lies on the graph of each equation in the xy -plane for the given system?

A. $(\frac{r}{5} + 7, -\frac{r}{5} + 35)$

B. $(-\frac{3r}{2} + \frac{7}{2}, r)$

C. $(r, \frac{2r}{3} + \frac{7}{3})$

D. $(r, -\frac{3r}{2} + \frac{7}{2})$

SAT 10

Math Module 3

Question # ID

SAT10 M 3.22 8c5e6702

A window repair specialist charges \$220 for the first two hours of repair plus an hourly fee for each additional hour. The total cost for 5 hours of repair is \$400. Which function f gives the total cost, in dollars, for x hours of repair, where $x \geq 2$?

A. $f(x) = 60x + 100$

B. $f(x) = 60x + 220$

C. $f(x) = 80x$

D. $f(x) = 80x + 220$